Chapter 7 study guide: the water cycle and watersheds

This guide covers Ms. Reyes's Chapter 7 for the Friday, November 6 exam (40 questions). She said to know the terms, the order of the cycle, what a watershed is, and the freshwater numbers. Source: your chapter notes only.

Key terms

Term	Meaning
Evaporation	Liquid water becomes vapor, driven by the sun's heat. Happens at any temperature, not only at boiling.
Transpiration	Plants release water vapor through their leaves.
Condensation	Water vapor cools and becomes liquid droplets. This forms clouds.
Precipitation	Water falls as rain, snow, sleet, or hail.
Runoff	Water that flows over land into streams and rivers.
Infiltration	Water soaking into the soil. It can recharge groundwater.
Residence time	How long a water molecule stays in one place.
Aquifer	Layer of rock or sediment that holds and moves groundwater.
Water table	Top of the zone where the ground is completely saturated.
Recharge	Water added to an aquifer by infiltration.
Watershed	Land area that drains to one outlet: a river, a lake, or the sea.
Divide	The ridge or high ground that forms a watershed's boundary.

Concepts

Where Earth's water is

About 97% of Earth's water is salt water in the oceans. About 2.5% is fresh water. Of that fresh water, about 69% is frozen in glaciers and ice caps, about 30% is groundwater, and less than 1% is surface water such as lakes, rivers, and swamps. Only a tiny share of fresh water is easy for people to use.

Example: a lake counts as surface water, so it sits in the "less than 1%" slice of the fresh water.

The cycle, in order

1. Evaporation
2. Transpiration
3. Condensation
4. Precipitation
5. Runoff
6. Infiltration

Water can follow many paths, so the cycle is not one fixed line. It has no starting point.

Example: rain (precipitation) lands on a lawn, soaks in (infiltration), and a tree pulls it up and releases it through its leaves (transpiration).

Residence time

Place	About how long
Atmosphere	9 days
Rivers	2 weeks
Ocean	3,200 years
Glaciers and ice caps	Thousands of years

Example: a molecule in a river is gone in about 2 weeks, but one in the ocean can stay for about 3,200 years.

Groundwater

An aquifer holds and moves groundwater. Infiltration adds water to it, which is recharge. The water table is the top of the fully saturated zone. It rises and falls and is not flat, because it follows the land a little.

Watersheds

A watershed drains to one outlet, and a divide marks its edge. Anything that happens inside it, such as fertilizer, pollution, or paving, can affect the water at the outlet. Paved surfaces cut infiltration and increase runoff, so streams can rise faster after storms.

Water budget from class: Runoff = Precipitation - Infiltration - Evaporation. This ignores plants and storage.

Example (numbers made up for practice): 100 units of precipitation, 30 infiltration, and 50 evaporation leave 100 - 30 - 50 = 20 units of runoff.

Practice questions

1. Which fraction of Earth's water is fresh? *About 2.5%.*
2. Where is most fresh water? *About 69% is frozen in glaciers and ice caps.*
3. How is transpiration different from evaporation? *Transpiration comes from living plants through their leaves. Evaporation comes from surfaces.*
4. What forms clouds? *Condensation: vapor cools into liquid droplets.*
5. Which residence time is longest: atmosphere, rivers, or ocean? *Ocean, about 3,200 years.*
6. What is recharge? *Water added to an aquifer by infiltration.*
7. Why do streams rise faster after a storm in a paved watershed? *Pavement cuts infiltration and increases runoff.*
8. Using the class budget, what is runoff if precipitation is 80, infiltration is 25, and evaporation is 35? *80 - 25 - 35 = 20 (computed).*

Common mistakes

- Mixing up transpiration (plants) and evaporation (surfaces).
- Saying the water table is flat. It follows the land a little.
- Thinking the water cycle starts at the ocean. It has no starting point.
- Forgetting that fresh water is only about 2.5%, not "most" of Earth's water.

Next step: cover the right column of the Key terms table and the practice answers, and quiz yourself before Friday.